

System/App Overview

1. Role Management Subsystem

- This subsystem at this point in time of designing will have 2 actors (Admin & the H.O.D) where an admin REGISTERS all the actors in play (H.O.Ds, Coordinators, Lecturers and Students). The H.O.D is the one responsible for assigning the modules to the coordinators and the Lecturers, but the admin is the one who assigns modules to the students.

2. Authentication Management Subsystem

- All actors are in play here because this is the subsystem for LOGGING, authentication and updating applicable user information.

3. Attendance Management Subsystem

- Here at this point of the design, for this subsystem the lecturer and the student are in play. The Lecturer logs in, then navigates to the attendance section via the UI, (login -> select module->select attendance section). Then on the attendance section UI, the lecturer selects the module there are about to generate the attendance code for, the type of class it is (tutorial, practical, theory). Then clicks the button to generate the random 8 character code. Then the lecturer displays the code to the students in class “physical/in real life”. So the students can also navigate to their attendance section and input the code, for verification. Once the system has verified, the UI will display Attendance recorded and saved to the student.

4. Assessments Insertion Subsystem

- At this point of the design this subsystem will have the lecturer and student in play. Where the lecturer, for the module in question enters all the necessary assessment's(s) details that all the students will do, (assessment-name, due-date, overall mark/what it's out of and weight, that is what it contributes to the final semester pass mark, 5% or 10% etc). The student can just come and view this section to maybe see how much work is ahead and what assessments will make or break the semester.

5. Marks Management Subsystem

- At this point in time of the design, again here the lecturer and the student are in play. Here the Lecturer enters the marks and while entering the mark, they also indicate where the if the student submitted the assessment or not, because these two along with attendance, will be used to determine the students flag as at risk. A Lecturer can also have the option to update a student's marks later on if need be. A student can also view their marks after they have been uploaded by the lecturer.

6. Risk Evaluation Subsystem

- This subsystem has the Coordinators in play, after the system has done the calculations/math to assess which students are at risk, using a defined metric to identify the risk levels that are "high-risk", "moderate-risk", "low-risk". All the information is displayed on the UI for the coordinator in almost like a long list with information, such as student numbers, averages of the students in that particular module, there are coordinating, and the risk levels, with color-coordination, High-risk being red, moderate-risk being yellow/orange, low risk being green. The UI will also allow them to filter students according to risk, for

whatever reason, also there can click a student and see more of their information/performance.

7. Intervention Subsystem

On this subsystem at this time of the design, there are 2 actors, the H.O.D and the Coordinator. Here after the coordinator has identified a/the student(s) at high risk and decide there is a need to intervene. They write what we have decided to call an “intervention report”, where there are suggesting ways to help the student(s). After some time (maybe a month) has passed the coordinator will write what we have called the “follow-up report”, that will detail if the student has improved or not. The H.O.D can view both the intervention and follow-up report.

8. Performance Subsystem

This system just means the students are able to view their personalized information on their performance/academic trends in some form of graphs on a dashboard of sorts (they might be a better way to represent that does not need graphs but I do not know yet). Another question is what to use for the graphs, if we are indeed using them, marks, submissions, attendance, averages, in monthly time intervals?